

Push Tasks — Answers



1. Order 3: angle of rotation?

120°

2. Order 8: angle of rotation?

45°

3. 60° rotation: order?

Order 6

4. 120° rotation: order?

Order 3

5. 36° rotation: order?

Order 10

6. 4 lines of symmetry: infer?

Order is likely 4 (regular)

7. 5 lines of symmetry: infer?

Order is likely 5 (regular)

8. 6 lines of symmetry: infer?

Order is likely 6 (regular)

9. Relation for regular polygons?

Lines of symmetry = order of rotational symmetry = n

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10. Order 4, 0 lines:
possible?

**Yes — a swastika-like
shape**

11. Order 2, 0 lines:
example?

A parallelogram

12. Logo with order 3:
describe?

**Three-fold symmetric
design**

13. Logo with order 6:
describe?

**Six-fold symmetric
design**

14. Dodecagon order
12?

**Yes, regular 12-gon
has order 12**

15. 15-gon: angle of
rotation?

24°

16. 18-gon: angle of
rotation?

20°

17. Formula for angle
of rotation?

**360° ÷ n for regular
n-gon**

18. Matches at
72°, 144°, 216°, 288°:
order?

Order 5

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19. Matches at $40^\circ, 80^\circ, 120^\circ, 160^\circ, 200^\circ, 240^\circ, 280^\circ$: order?
Order 9

20. Matches at $90^\circ, 180^\circ, 270^\circ$: order?
Order 4

21. Two shapes same order: example?
Square and rhombus (both order 4 and 2)

22. Rotational but no reflection: example?
Parallelogram (order 2, no reflection)

23. Reflection but no rotational: example?
Isosceles triangle (order 1, 1 line)

24. Traffic sign: order?
Stop sign: order 8

25. Car logo: order?
Mitsubishi: order 3

26. Flag with order 2?
UK flag: order 2

27. Order > sides: possible?
No — order can't exceed sides for polygon